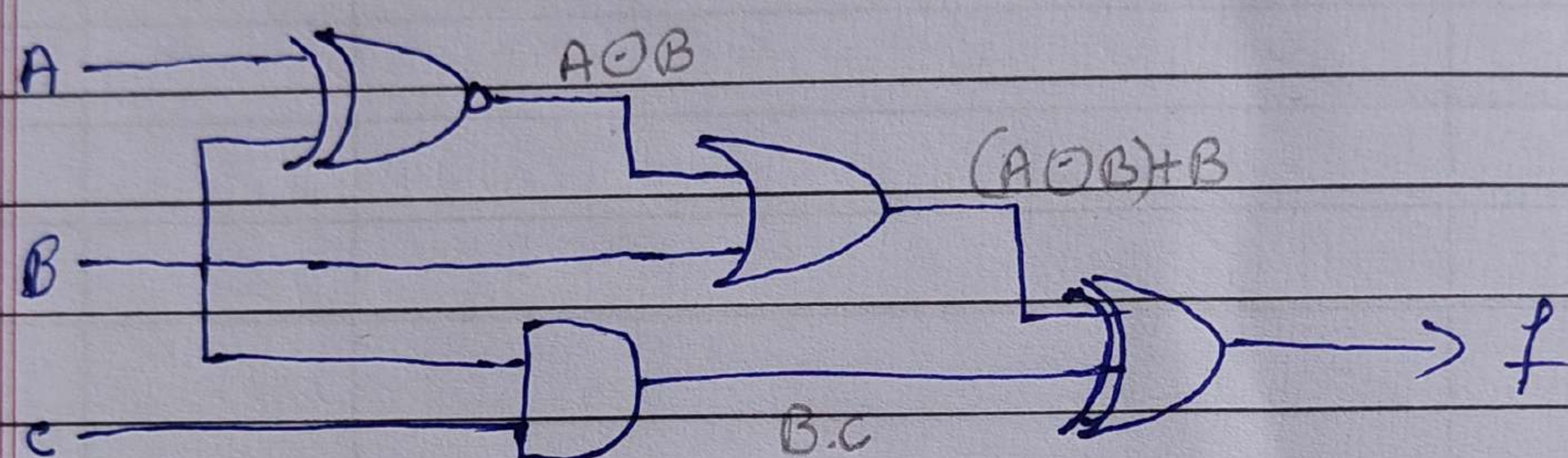
1. Derive the output expression 'f' for the below circuits and draw the simplified circuits.

a)



$$f = (A \odot B) + B \oplus B.C$$

$$= AB + \bar{A}\bar{B} + B \oplus BC$$

$$= B(A+1) + \bar{A}\bar{B} \oplus BC$$

$$= B + \bar{B}\bar{A} \oplus BC$$

$$= \underbrace{\cancel{(B+\bar{B})}}_{1}(B+\bar{A}) \oplus BC$$

$$= (B+\bar{A}) \oplus BC$$

$$= (B+\bar{A})\overline{BC} + \overline{(B+\bar{A})}BC$$

$$= (B+\bar{A})(\bar{B}+\bar{C}) + \underbrace{\bar{B}.A BC}_{0}$$

$$= 0 + B\bar{C} + \bar{A}\bar{B} + \bar{A}\bar{C}$$

$\cancel{\bar{A}\bar{C} + \bar{A}\bar{B} +}$

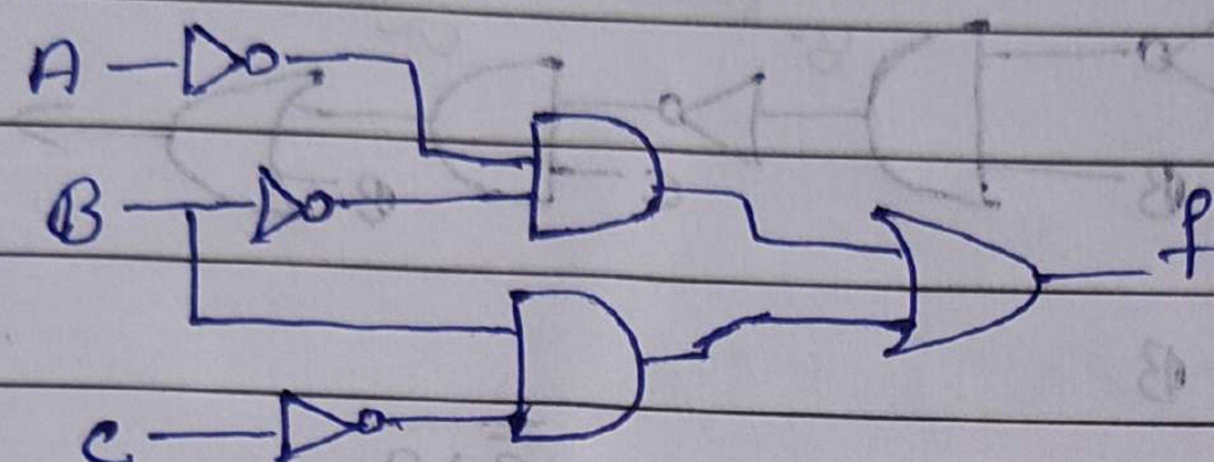
$$= B\bar{C} + \bar{B}\bar{A} + \bar{C}\bar{A}$$ Extra term

$$= B\bar{C} + \bar{B}\bar{A} + \bar{C}\bar{A}(B+\bar{B})$$

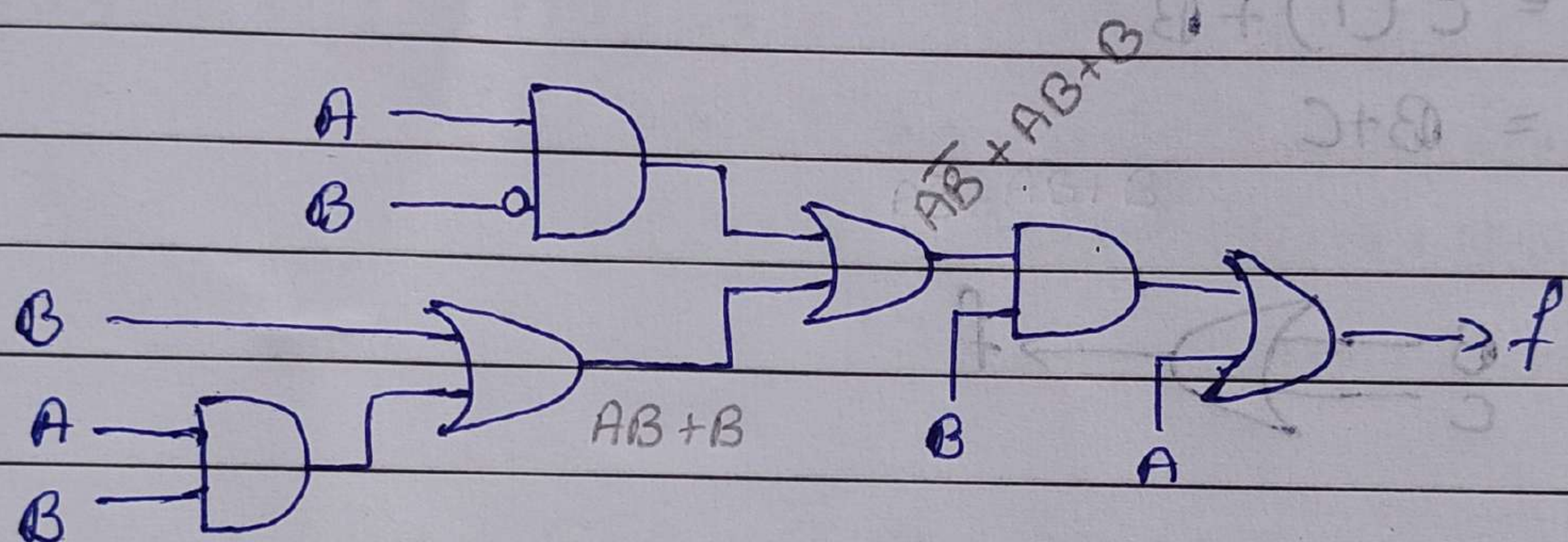
$$= B\bar{C} + \bar{B}\bar{A} + \bar{C}\bar{A}B + \bar{C}\bar{A}\bar{B}$$

$$= B\bar{C}(1+\bar{A}) + \bar{B}\bar{A}(1+\bar{C})$$

$$f = B\bar{C} + \bar{B}\bar{A}$$



b)



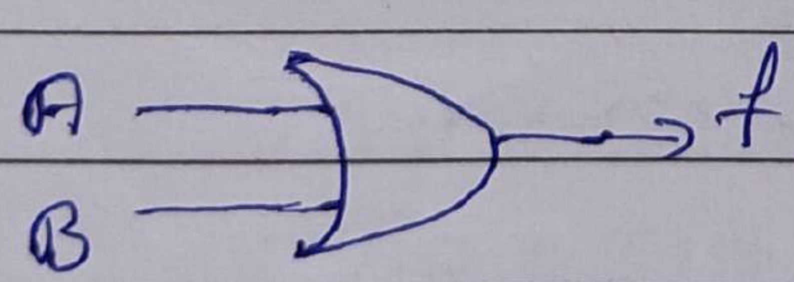
$f = (A\overline{B} + AB + B).B + A$

$= \cancel{A\overline{B}B}^{\,0} + ABB + BB + A$

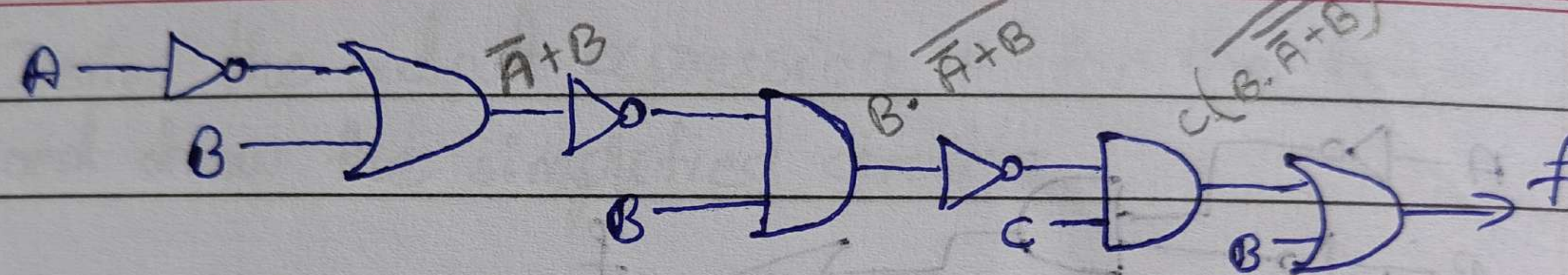
$= AB + B + A$

$= A(B+1) + B$

$f = A+B$



c)



$$f = C\left(\overline{B.(\overline{\overline{A}+B})}\right) + B$$

$$= C\left(\overline{B.A\overline{B}}\right) + B$$

$$= C(\overline{0}) + B$$

$$= C(1) + B$$

$$f = B + C$$

$$= \overline{\overline{A}+B}$$

$$= A\overline{B}$$

2.

2. Minimize the following Boolean Expressions using Boolean postulates.

a) $\bar{A}C\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

$= \bar{A}C\bar{D}E(B+\bar{B}) + \bar{A}\bar{B}\bar{D} + ABCE(D+\bar{D}) + ABD$

$= \bar{A}C\bar{D}EB + \bar{A}C\bar{D}E\bar{B} + \bar{A}\bar{B}\bar{D} + ABCDE + ABCE\bar{D} + ABD$

$= \bar{A}\bar{B}\bar{D}(1+CE) + \bar{A}C\bar{D}EB + ABD(1+\cancel{B}CE) + ABCE\bar{D}$

$= \bar{A}\bar{B}\bar{D} + \bar{A}C\bar{D}EB + ABD + ABCE\bar{D}$

$= ABD + \bar{A}\bar{B}\bar{D} + BC\bar{D}E(A+\bar{A})$

$= ABD + \bar{A}\bar{B}\bar{D} + BC\bar{D}E$

b) $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

$\bar{B}\bar{C}(\bar{A}+A) + ABD + \bar{A}C(1+\bar{D}) + A\bar{C}D$

$\bar{B}\bar{C} + ABD + \bar{A}C + A\bar{C}D$

$BAD + \bar{B}\bar{C} + AD\bar{C} + \bar{A}C$ Redundant

$= ABD + \bar{B}\bar{C} + \bar{A}C$

c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + AC\bar{D} + \bar{B}\bar{C}\bar{D}$

$\bar{A}\bar{C} + BC(1+\bar{A}) + A\bar{B} + AC\bar{D} + \bar{B}\bar{C}\bar{D}$

$\bar{A}\bar{C} + BC + A\bar{B} + AC\bar{D} + \bar{B}\bar{C}\bar{D}$

$\bar{A}\bar{C} + \bar{B}\bar{C}\bar{D}A + \bar{B}\bar{C}\bar{D}\bar{A} + BC + A\bar{B} + AC\bar{D}B + AC\bar{D}\bar{B}$

$\bar{A}\bar{C}(1+\bar{B}\bar{D}) + \bar{B}\bar{C}\bar{D}A + BC(1+A\bar{D}) + A\bar{B}(1+C\bar{D})$

$\bar{A}\bar{C} + \bar{B}\bar{C}\bar{D}A + BC + A\bar{B}$

$\bar{A}\bar{C} + BC + A\bar{B}(1+\bar{C}\bar{D})$

$\bar{A}\bar{C} + BC + A\bar{B}$

d) $\bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_1x_2\bar{x}_3x_4 + x_2$

$\bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + \bar{x}_1\bar{x}_2x_4 + x_2(x_1\bar{x}_3x_4 + 1)$

$\bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + x_2 + \bar{x}_2\bar{x}_1x_4$

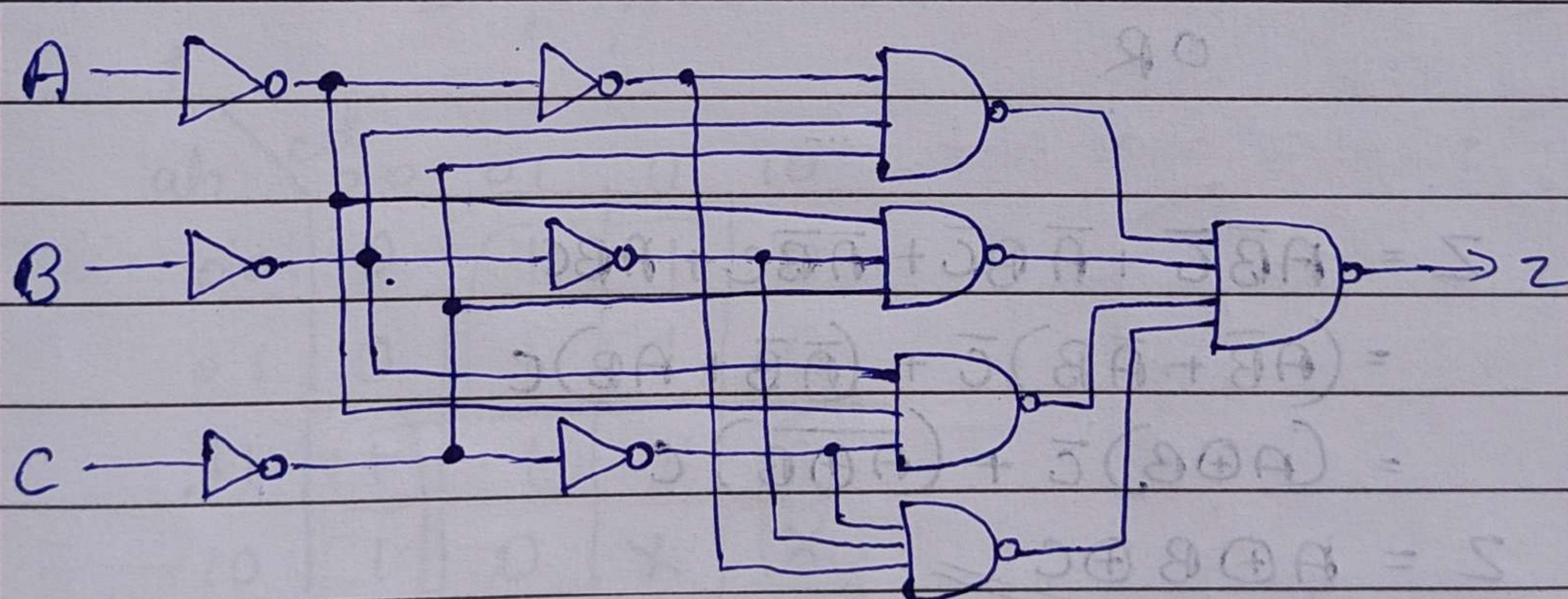
$\bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + x_2 + \bar{x}_1x_4$

$\bar{x}_1\bar{x}_3\bar{x}_4 + x_2 + x_3x_4 + \bar{x}_1x_3x_4 + \bar{x}_1\bar{x}_3x_4$

$\bar{x}_1\bar{x}_3(\bar{x}_4 + x_4) + x_2 + x_3x_4(1 + \bar{x}_1)$

$\bar{x}_1\bar{x}_3 + x_2 + x_3x_4$

3. Which single gate can represent the functionality of given circuit?



$\cancel{Z = A\bar{B}\bar{C} + \bar{A}B\bar{C}}$

$Z = \overline{\overline{A\bar{B}\bar{C}} \cdot \overline{\bar{A}B\bar{C}} \cdot \overline{\bar{A}\bar{B}C} \cdot \overline{ABC}}$

$Z = A\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C + ABC$

Truth table

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Z is high when odd number
∴ it represents XOR gate.

OR

$$Z = A\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C + ABC$$
$$= (A\bar{B} + \bar{A}B)\bar{C} + (\bar{A}\bar{B} + AB)C$$
$$= (A \oplus B)\bar{C} + (\overline{A \oplus B})C$$
$$Z = A \oplus B \oplus C$$

3 input XOR gate

4 Minimize the following Boolean Expressions using K-map

a) $f(a,b,c,d) = \sum_m (4,6,8,10,11,12,15) + \sum_d (3,5,7,9)$

ab \ cd	00	01	11	10
00	$_0$	$_1$	X $_3$	$_2$
01	1 $_4$	X $_5$	X $_7$	1 $_6$
11	1 $_{12}$	$_{13}$	1 $_{15}$	$_{14}$
10	1 $_8$	X $_9$	1 $_{11}$	1 $_{10}$

$f = \bar{a}b + a\bar{b} + cd + b\bar{c}\bar{d}$ (or $a\bar{c}\bar{b}$)

b) $f(a,b,c,d) = \prod M(0,2,4,6,7,9) \cdot \prod D(10,11)$

f

ab \ cd	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	1	1	1
10	1	0	X	X

$f = \bar{a}d + ab + a\bar{d}$